Model MEML 75:

A phenomenological parametrization of the Standard Model quark sector from $T^2/\mathbb{Z}_3$ geometry with axion-like (ALP) dark matter

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Abstract

We present model MEML 75, which parametrizes the quark sector of the Standard Model through the geometry of the six-dimensional orbifold $T^2/\mathbb{Z}_3$ with two independent tori. The model fits **9 quark observables** (5 masses + 3 CKM + J) with **11 free parameters** ($N_{\rm dof} = 9 - 11 = -2$ in the quark sector).

The main testable prediction: axion-like dark matter with $f_a = 1.9801 \times 10^{11}$ GeV, $m_a = 28.407$ μeV, $\nu_a = 6.8688$ GHz, and ALP-photon coupling $g_{a\gamma\gamma} = +4.38 \times 10^{-15}$ GeV^{-1} (DFSZ) or -1.13×10^{-14} GeV^{-1} (KSVZ), testable at ADMX/HAYSTAC (0.5–10 GHz). The relic density $\Omega_a h^2$ is reproduced with one free parameter θ_i.

Classification of quantities: $\alpha_G = 2/(3\pi)$ is a *postulate* with geometric motivation; $\varepsilon = (\alpha_G/(4\pi))^2$ is a one-loop *ansatz*; m_{KK}, M_6 are conditionally derived at $\mathcal{I} = 0.3653$.

Quark sector. A global fit with loss function $\chi^2 = \sum_i ((O_i - O_i^{\rm target})/\sigma_i)^2$ yields $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1577944085 + 2.3851782590\,i$, $\tau_d = +0.3693905617 + 1.5410496057\,i$. Independent reproduction is given in §8 and Appendix C. $\mathrm{Im}\,\tau$ is stable ($\pm 0.3\%$), $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).

Systematic tests (all negative): (1) the universal Yukawa ansatz is excluded; (2) 36 matrix structures are equally unstable (spread of $|V_{ub}|$ from -97% to $+25407\%$); (3) θ_i from inflation is excluded at 26σ; (4) modular forms at $\tau = e^{i\pi/3}$ do not yield α_G exactly ($|Y_2^{(6)}|^2 = 0.445302 \neq 4/9$).

Coefficient degeneracy. At $N_{\rm dof} = 9 - 11 = -2$ the coefficients c_i are non-unique: χ^2 is invariant under $c_i \to k c_i$ (Appendix D). The model *interpolates* quark observations, it does not predict them.

Verdict: the model is *phenomenological*.

1 Introduction

1.1 Motivation

Modular models of fermion masses [4, 5] build the mass hierarchy and mixing from compactification geometry. In model MEML 75, quarks are localized on two independent tori $T^2/\mathbb{Z}_3$ with moduli τ_u and τ_d, and the mass matrices are built from A_4 modular forms of weights $(2, 4, 6)$. An axion-like particle (ALP) arising from the same geometric ingredients provides dark matter. The lepton sector is deferred to future work.

1.2 Self-containment

The paper is constructed so that the reader can reproduce *every* number without external sources:

- all Y-forms (q-expansion coefficients) — in Appendix A;
- all PDG target values — in §8;
- DE-optimization procedure with parameters — in Appendix B;
- three independent fits — in Appendix C;
- proof of scale-degeneracy — in Appendix D;
- flatness procedure — in Appendix E;
- LOO procedure — in Appendix F;
- systematic search for α_G (392 combinations) — in Appendix G.

1.3 Conventions

Table 1: Status of quantities.

Status	Meaning
[POSTULATE]	Postulate with geometric motivation
[ANSATZ]	Phenomenological ansatz
[COND]	Conditionally derived (at $\mathcal{I}$)
[MATCH]	Structural matching relation
[DERIVED]	Algebraic consequence
[FIT]	From numerical fit
[OBSERVED]	Experimental value

2 Geometry of $T^2/\mathbb{Z}_3$

2.1 Metric and modular parameter

The orbifold $T^2/\mathbb{Z}_3$:

$$z = x + \tau y, \quad ds^2 = dx^2 + dx\,dy + dy^2, \tag{2.1}$$

where $\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$.

2.2 Fixed points

Z_3-action $z \to \omega z$, $\omega = e^{2\pi i/3}$:

$$z_1 = (0,0), \quad z_2 = (\tfrac{1}{3}, \tfrac{1}{3}), \quad z_3 = (\tfrac{2}{3}, \tfrac{2}{3}). \tag{2.2}$$

These three points provide a topological explanation for three generations.

2.3 Two-torus structure

Quarks are localized on different tori with moduli:

$$\boxed{\tau_u = -0.1577944085 + 2.3851782590\,i, \quad \tau_d = +0.3693905617 + 1.5410496057\,i} \tag{2.3}$$

Status: [FIT]. Determined by the global fit (§8, Appendix B). The values are given to 10 significant digits because the minimum is razor-thin: rounding to 3 significant digits raises χ^2 from 6×10^{-20} to 2.6×10^{-3} (Appendix C). $\mathrm{Im}\,\tau_u = 2.385$, $\mathrm{Im}\,\tau_d = 1.541$ are stable across all fits; $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).

3 Bulk coupling α_G (postulate)

3.1 Definition

$$\boxed{\alpha_G = \frac{2}{3\pi} = 0.2122065908} \tag{3.1}$$

Status: [POSTULATE]. Geometric motivation: 2/3 is the volume fraction of the fundamental domain of the $\mathbb{Z}_3$-orbifold; $1/\pi$ is the normalization of the theta-function period.

3.2 Algebraic identity

$$\frac{2\pi}{3\alpha_G} = \pi^2 = 9.8696044011. \tag{3.2}$$

This is an identity, not a coincidence. Numerically: $e^{\pi^2} = 1.933369 \times 10^4$.

3.3 Check of modular forms at $\tau = e^{i\pi/3}$

At $\tau = e^{i\pi/3}$ (60 digits, 30 terms, Appendix A):

Table 2: A_4 modular forms at $\tau = e^{i\pi/3}$.

Form	Modulus	Status
$\lvert Y_1^{(2)}\rvert$	0.948674	—
$\lvert Y_2^{(2)}\rvert$	0.974090	—
$\lvert Y_3^{(2)}\rvert$	0.476383	—
$\lvert Y_1^{(4)}\rvert$	1.364023	—
$\lvert Y_2^{(4)}\rvert$	0.697153	—
$\lvert Y_3^{(4)}\rvert$	1.400784	—
$\lvert Y_1^{(6)}\rvert$	0.297141	—
$\lvert Y_2^{(6)}\rvert$	0.667310	—
$\lvert Y_3^{(6)}\rvert$	0.679090	—

Checks of α_G:

- $|Y_2^{(6)}|/\pi = 0.212411$ (deviation $+0.0965\%$);
- $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (deviation -0.0385%).

The exact equality $|Y_2^{(6)}| = 2/3$ *does not hold*: $|Y_2^{(6)}|^2 = 0.445302 \neq 4/9 = 0.444444$, $|\Delta| = 8.576 \times 10^{-4}$, stable at $n = 5, 11, 20, 30$.

3.4 Systematic search for a derivation of α_G

We performed an exhaustive search over the basis $\mathcal{B} = \{Y_i^{(2)}, Y_i^{(4)}, Y_i^{(6)}, \eta, \pi, \sqrt{\pi}, \pi^2, e^{-\pi^2}\}_{i=1,2,3}$ at $\tau = e^{i\pi/3}$. Here $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind eta function, $q = e^{2\pi i\tau}$; $|\eta(e^{i\pi/3})| = 0.800579$.

Tested $N = 392$ combinations. Precision: 30 terms, 60-digit arithmetic.

Found 2 coincidences below 0.1%:

Table 3: Structural near-coincidences.

Combination	Value	Deviation	Stability
$\lvert Y_1^{(6)}\rvert/\lvert Y_3^{(4)}\rvert$	0.212125	$\mathbf{-0.0385}\%$	$n = 5..30$
$\lvert Y_2^{(6)}\rvert/\pi$	0.212411	$\mathbf{+0.0965}\%$	$n = 5..30$

Statistical estimate. The expected number of hits at $N = 392$ with tolerance $\pm 0.1\%$: $\langle n\rangle = 392 \cdot 0.002 = 0.78$. Observed $n_{\text{obs}} = 2$. Poisson probability:

$$P(n \geq 2 \mid \langle n\rangle = 0.78) = \frac{\gamma(2, 0.78)}{\Gamma(2)} = 0.185. \tag{3.3}$$

This is **not significant** ($p = 0.185 > 0.05$).

Conclusion. $\alpha_G = 2/(3\pi)$ remains a **postulate**.

4 Suppression factor ε (ansatz)

$$\boxed{\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{1}{36\pi^4} = 2.851662 \times 10^{-4}} \tag{4.1}$$

Status: [ANSATZ]. Motivation: one-loop structure of QFT.

4.1 Matching 6D↔4D

Holomorphic Yukawa coupling: $Y_{\text{hol}}(\tau) = \sum_n N_n e^{-2\pi n \operatorname{Im}\tau} P_n(\tau)$. After canonical normalization:

$$Y_{\text{phys}} \sim e^{-S_{\text{inst}}}\,(\operatorname{Im}\tau)^{\Delta k}, \quad \Delta k = \tfrac{k_i+k_j+k_k}{2} - b, \tag{4.2}$$

with $\Delta k = 1$ (up), $\Delta k = 2$ (down).

Numerical check: $e^{-\pi^2} = 5.1723 \times 10^{-5}$; $\varepsilon/e^{-\pi^2} = 5.514$; $(2\operatorname{Im}\tau_u)^1 = 4.774$; $(2\operatorname{Im}\tau_d)^2 = 6.248$. $5.514 \in [4.774, 6.248]$ — within range.

5 Conditionally derived quantities

$$m_{KK} = \frac{M_{Pl}^{\rm red}}{\sqrt{\mathcal{I}}\, e^{\pi^2}} = 2.0838 \times 10^{14}\ \text{GeV} \quad \text{[COND]}, \tag{5.1}$$

$$M_6 = \left(3(M_{Pl}^{\rm red})^2 m_{KK}^2\right)^{1/4} = 2.9646 \times 10^{16}\ \text{GeV} \quad \text{[COND]}, \tag{5.2}$$

at $\mathcal{I} = 0.3653$. $M_6 \propto \mathcal{I}^{-1/4}$. For instance, at $\mathcal{I} = 0.1$, the dynamically calculated scale is $M_6 = 4.098 \times 10^{16}$ GeV, which rounds to 4.10×10^{16} GeV.

6 Axion-like (ALP) sector

6.1 ALP parameters

$$f_a = M_{Pl}^{\rm red}\, \varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{6.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = \frac{(75.0\ \text{MeV})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{6.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{6.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD, which is the correct quantity entering the axion mass formula (not the QCD scale $\Lambda_{\rm QCD} \approx 24$ MeV).[1]

We emphasize that no explicit $U(1)_{PQ}$ symmetry is introduced in this model. The parameter E/N is a discrete choice rather than a theoretical prediction. Furthermore, the axion quality problem is not addressed within this framework.

Status: [COND]. ν_a within ADMX/HAYSTAC range.

6.2 ALP-photon coupling

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm EM}}{2\pi f_a}\left(\frac{E}{N} - 1.92\right). \tag{6.4}$$

At $f_a = 1.9801 \times 10^{11}$ GeV: $\alpha_{\rm EM}/(2\pi f_a) = 5.866 \times 10^{-15}\ \text{GeV}^{-1}$.

Table 4: Prediction for $g_{a\gamma\gamma}$.

Scenario	E/N	$g_{a\gamma\gamma}$, GeV^{-1}
KSVZ	0	-1.126×10^{-14}
DFSZ	8/3	$+4.379 \times 10^{-15}$

Both values lie within the sensitivity of ADMX Run 1D–1E. The factor of 2.6 between them means that the model predicts a *two-point* target.

[1] Using $\chi^{1/4} = 75.5$ MeV would give $m_a = 28.8\ \mu$eV and $\nu_a = 6.96$ GHz; both values lie within the theoretical uncertainty of $\chi^{1/4}$ and are testable at ADMX/HAYSTAC.

7 A remark on $SU(5)$

While $SU(5)$ is the unique rank-4 group admitting the required SM content, the quark sector here is built on A_4 modular forms. The $SU(5)$ structure is not used for the mass generation mechanism and is mentioned only for completeness.

8 Quark sector (11 parameters)

We assume that the Yukawa couplings are defined at the GUT scale. RG running down to the electroweak scale is neglected, following the standard practice in modular flavor models [4]; it introduces $\mathcal{O}(1)$ uncertainties but does not affect the structural findings (coefficient degeneracy, LOO instability).

8.1 Matrix structure

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix}, \; M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{8.1}$$

8.2 Loss function and target values

$$\chi^2 = \sum_{i=1}^{9} \left(\frac{O_i(\tau_u, \tau_d, c) - O_i^{\text{target}}}{\sigma_i} \right)^2. \tag{8.2}$$

Table 5: Target values (PDG 2024).

Observable	$O_i^{\text{target}} \pm \sigma_i$
m_u/m_t	$(1.27 \pm 0.30) \times 10^{-3}/172.0$
m_c/m_t	$(0.620 \pm 0.020)/172.0$
m_d/m_b	$(2.70 \pm 0.20) \times 10^{-3}/2.860$
m_s/m_b	$(55.0 \pm 5.0) \times 10^{-3}/2.860$
m_b/m_t	$(2.860 \pm 0.030)/172.0$
$\lvert V_{us}\rvert$	0.22501 ± 0.00068
$\lvert V_{cb}\rvert$	0.04053 ± 0.00042
$\lvert V_{ub}\rvert$	0.00382 ± 0.00020
$\lvert J\rvert$	$(3.080 \pm 0.130) \times 10^{-5}$

8.3 Results

Independent optimization (DE, 3 seeds, Appendix B):

$$\boxed{\chi^2 = 6.195 \times 10^{-20} \quad \text{at} \quad \tau_u = -0.1577944085 + 2.3851782590\,i, \; \tau_d = +0.3693905617 + 1.5410496057} \tag{8.3}$$

Coefficients:

$$c^{\text{fit}} \approx (5.31 \times 10^{-5},\ 7.19,\ 0.039,\ 2.43 \times 10^{-4},\ 0.503,\ 1.37 \times 10^{-3},\ 2.22 \times 10^{-3}). \tag{8.4}$$

Per-observable pulls at these values: all $< 10^{-9}\sigma$.

8.4 Independent reproduction

Three independent 11-parameter DE runs (Appendix C) converge to $\mathrm{Im}\,\tau_u = 2.385$, $\mathrm{Im}\,\tau_d = 1.541$ to 0.3%, with $\chi^2 < 10^{-18}$ in all runs.

8.5 Stability of CKM under shifts in τ

With coefficients (8.4) held fixed, a shift in τ_u or τ_d by $\pm 5\%$ moves the CKM elements outside 1σ: $\mathrm{Im}\,\tau_u = 2.385 \pm 0.007$, $\mathrm{Im}\,\tau_d = 1.541 \pm 0.005$; $\mathrm{Re}\,\tau_d \in [-0.37, +0.37]$ (soft flatness, Appendix E); $\mathrm{Re}\,\tau_u \in [-0.16, -0.13]$.

Consequence for flux. Only $\mathrm{Im}\,\tau$ requires precise stabilization ($\pm 3\%$).

9 Test B.1: universal ansatz

Table 6: Levels of coefficient freedom.

Level	Parametrization	k_{coef}	N_{dof}	χ^2
L1	$a_u = b_u = c_u = y_u$	2	+3	5.28×10^6
L2	$+r$	3	+2	3.51×10^2
L3	$+r_u$	4	+1	4.45×10^3
L4	up + 1 ratio down	5	0	3.53×10^3
L5	2 ratios per sector	6	−1	4.36×10^2
L6	**7 free**	**7**	**−2**	$\mathbf{2.18 \times 10^{-19}}$

Conclusion. The universal ansatz is **excluded**. Only L6 yields the ideal fit.

10 Test C.3: scan of matrix structures

We tested 36 combinations (6 up × 6 down) with 6 seeds.

Table 7: LOO for U6×D6.

Seed	χ^2	$\lvert V_{ub}\rvert$ err	$\lvert J\rvert$ err
0	8.16×10^{-9}	+25406.7%	−97.2%
1	2.09×10^{-8}	+24308.0%	+20318.7%
2	1.33×10^{-8}	+33.1%	−97.5%
3	1.04×10^{-10}	+8.4%	+19.2%
4	9.41×10^{-10}	+25302.9%	+872.6%
5	1.43×10^{-9}	+25311.8%	+570.6%

Conclusion. All 36 structures are unstable. The problem lies in over-determination ($N_{\mathrm{dof}} = -2$).

11 ALP dark matter

11.1 Analytical formula

$$\Omega_a h^2 \approx 0.12\,\theta_i^2\, f_{\mathrm{anh}}(\theta_i) \left(\frac{f_a}{3 \times 10^{11}}\right)^{7/6}, \quad f_{\mathrm{anh}}(\theta_i) = [\ln(e/(1-(\theta_i/\pi)^2))]^{7/6}. \tag{11.1}$$

11.2 Full decomposition

Step 1 (bare): $\Omega_a h^2_{\rm bare} = 0.12 \cdot 1.2459 \cdot 0.6159 = 0.0921$.

Step 2 (Lyth–Turner anharmonicity): $f_{\rm anh}(1.1162) = 1.1591$, $\Omega_a h^2_{\rm anh} = 0.1067$.

Step 3 (T-effects): $\Omega_a h^2_{\rm num} = 0.1067 \cdot (1 + \delta_T)$, $\delta_T = +0.124$. $\delta_T = +12.4\%$: $g_*(T_{\rm osc})$ (+5%), $T_{\rm osc}/\Lambda_{\rm QCD}$ (+4%), $m_a(T)$ (+3%).

Result. Numerical vs analytical gap: +12.4% (not +57%). Agreement with Planck at $\theta_i = 1.1162$ requires no tuning.

12 Test B.2: θ_i from inflation

$\delta\theta = H_{\rm inf}/(2\pi f_a)$, $\beta_{\rm iso} = (\delta\theta/\theta_i)^2$. Constraints: $r < 0.036 \Rightarrow H < 4.703 \times 10^{13}$ GeV; $\beta_{\rm iso} < 0.038 \Rightarrow H < 2.707 \times 10^{11}$ GeV.

Hypothesis $\theta_i \sim \delta\theta$ requires $H = 1.389 \times 10^{12}$ GeV, but $\beta_{\rm iso} = 1.0$ — **excluded at** 26σ. $\theta_i = 1.1162$ is a random draw from $[-\pi, \pi]$.

13 Test A.1: modular forms at $\tau = e^{i\pi/3}$

Details in §3.3, §3.4. Summary:

Table 8: Combinations of forms and deviation from $\alpha_G = 0.212207$.

Combination	Value	Deviation
$\lvert Y_1^{(6)}\rvert/\lvert Y_3^{(4)}\rvert$	0.212125	−0.0385%
$\lvert Y_2^{(6)}\rvert/\pi$	0.212411	+0.0965%
$\lvert Y_3^{(6)}\rvert/\pi$	0.216161	+1.86%
$\lvert Y_2^{(4)}\rvert/\pi$	0.221911	+4.57%

$|Y_2^{(6)}|^2 = 0.445302 \neq 4/9$; stable at $n = 5..30$. PSLQ finds nothing. p-value = 0.185 (not significant).

Conclusion. α_G remains a postulate.

14 Full balance

Table 9: Free parameters (13).

Category	Parameters	Count
Fermion moduli	τ_u, τ_d (complex)	4
up coefficients	a_u, b_u, c_u	3
down coefficients	a_1, a_2, b, c	4
Topological	$\mathcal{I}$	1
ALP	θ_i	1
Total		**13**

Table 10: Observables: fitted, predicted, and future work.

Sector	Observables	Status
Quark masses	$m_u/m_t, \ldots, m_b/m_t$	5 (fitted)
Quark CKM	$\lvert V_{us}\rvert, \lvert V_{cb}\rvert, \lvert V_{ub}\rvert$	3 (fitted)
Quark (J)	$\lvert J\rvert$	1 (fitted)
ALP	ν_a, $g_{a\gamma\gamma}$	predicted
Cosmology	$\Omega_a h^2$	1 (fitted via θ_i)
Lepton PMNS	$\theta_{12}, \theta_{13}, \theta_{23}, \lvert J\rvert$	4 (future work)
Fitted quarks		**9**

$$N_{\text{dof}}^{\text{quarks}} = 9 - 11 = -2. \tag{14.1}$$

14.1 Information criteria

Table 11: AIC/BIC for the quark sector ($n = 9$). Here k_q is the number of quark-sector parameters.

Model	k_q	χ^2	AIC	BIC
MEML 75	11	$\approx 10^{-20}$	22.0	24.2
SM	10	0.9	20.9	22.9

A strict AIC/BIC comparison for the quark sector alone does not show a decisive statistical preference over the Standard Model ($k_q = 11$ for our model vs. $k_q^{SM} = 10$ for the SM). We therefore present our model primarily as a phenomenological parametrization of quark parameters from $T^2/\mathbb{Z}_3$ geometry, rather than as a statistically preferred alternative.

15 Open problems

1. **Coefficient degeneracy.** At fixed τ (2.3), different DE seeds find coefficients with $\chi^2 < 10^{-18}$, related by uniform scale $\times 1.87$. Check: $\chi^2(c) = \chi^2(kc) < 10^{-19}$ for any $k \in [0.5, 10]$ (Appendix D). Direct consequence of $N_{\text{dof}} = -2$. Priority for v8.0: mechanism to fix c_i (holomorphic 6D→4D couplings).

2. **Status of α_G.** Postulate. Systematic search yields no exact formula; near-coincidences are statistically expected ($p = 0.185$).

3. **Status of ε.** Ansatz; choice of 4π is not derived.

4. **Flatness of $\operatorname{Re}\tau$.** $\operatorname{Re}\tau_d$ is soft-flat (width 0.74); $\operatorname{Re}\tau_u$ is soft-flat (width 0.03). $\operatorname{Im}\tau$ is stable.

5. **Matching 6D↔4D.** Structural.

6. **LOO instability.** At $N_{\text{dof}} = -2$, spread of $\lvert V_{ub}\rvert$ from -97% to $+25407\%$.

7. **Over-determination in quarks.** 11 parameters for 9 observables.

8. **Universal ansatz excluded.** Only L6 yields ideal χ^2.

9. **GUT dependence on $\mathcal{I}$.** $M_6 \propto \mathcal{I}^{-1/4}$.

10. **Analytic vs numerical $\Omega_a h^2$.** $+12.4\%$, explained by T-effects.

11. **Derivation of θ_i.** Excluded at 26σ.

16 Roadmap

Table 12: Phases of development.

Phase	Task	Priority
I	Mechanism for c_i: holomorphic 6D→4D couplings	Critical
I	Flux stabilization of $\mathrm{Im}\,\tau$ ($\pm 3\%$)	High
II	Reduce $11 \to 9$ in quarks	Medium
II	Analytical derivation of c_i	Medium
III	ADMX, DESI, DUNE, KamLAND-Zen	External
IV	Leptogenesis, dark energy	Long-term

17 Conclusions

What has been done:

1. A quark-sector fit with 11 parameters for 9 observables ($N_{\mathrm{dof}} = -2$), plus a testable ALP prediction.
2. 4 systematic tests (B.1, C.3, B.2, A.1) — all negative.
3. A global fit $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1578 + 2.3852i$, $\tau_d = +0.3694 + 1.5410i$; independently reproduced (3 DE runs).
4. A test of coefficient degeneracy: χ^2 is invariant under $c_i \to k c_i$.
5. $\Omega_a h^2$: bare → anh → num, $\delta_T = +12.4\%$.
6. $g_{a\gamma\gamma}$ (KSVZ/DFSZ) — two-point target for ADMX.
7. AIC/BIC with $N_{\mathrm{dof}} < 0$ caveat.

What works: CKM fit ($\chi^2 \sim 10^{-20}$), Planck compatibility of dark matter, testable prediction $\nu_a = 6.8688$ GHz, $g_{a\gamma\gamma} \in \{+4.38 \times 10^{-15}, -1.13 \times 10^{-14}\}$ GeV^{-1}.

What does not work: LOO is unstable; coefficients are scale-degenerate; α_G, ε, θ_i are not derived from first principles.

Verdict. The model is *phenomenological*. It reproduces observations but does not predict. The coefficient degeneracy shows that the model interpolates 9 observations with 11 parameters.

Priority for future work: a mechanism to fix c_i. Flux is only needed for $\mathrm{Im}\,\tau$ at $\pm 3\%$.

A A_4 modular forms: full q-expansions

Definition: $q = e^{2\pi i \tau}$.

$$Y_1^{(2)} = 1 + \sum_{n=1}^{17} C_1^{(n)} q^n, \tag{A.1}$$

$$Y_2^{(2)} = -6\, q^{1/3} \left(1 + \sum_{n=1}^{11} C_2^{(n)} q^n\right), \tag{A.2}$$

$$Y_3^{(2)} = -18\, q^{2/3} \left(1 + \sum_{n=1}^{10} C_3^{(n)} q^n\right), \tag{A.3}$$

$C_1 = [12, 36, 12, 84, 72, 36, 96, 180, 12, 216, 144, 36, 156, 96, 84, 240, 36]$; $C_2 = [1, 7, 8, 18, 14, 30, 20, 42, 30, 56, 42]$; $C_3 = [1, 2, 5, 6, 12, 8, 14, 10, 22, 16]$.

Convolutions:

$$Y_1^{(4)} = (Y_1^{(2)})^2 - Y_2^{(2)} Y_3^{(2)}, \tag{A.4}$$

$$Y_2^{(4)} = (Y_3^{(2)})^2 - Y_1^{(2)} Y_2^{(2)}, \tag{A.5}$$

$$Y_3^{(4)} = (Y_2^{(2)})^2 - Y_1^{(2)} Y_3^{(2)}, \tag{A.6}$$

$$Y_1^{(6)} = \tfrac{1}{3}\left[2Y_1^{(2)} Y_1^{(4)} - Y_2^{(2)} Y_3^{(4)} - Y_3^{(2)} Y_2^{(4)}\right], \tag{A.7}$$

$$Y_2^{(6)} = \tfrac{1}{3}\left[2Y_3^{(2)} Y_3^{(4)} - Y_1^{(2)} Y_2^{(4)} - Y_2^{(2)} Y_1^{(4)}\right], \tag{A.8}$$

$$Y_3^{(6)} = \tfrac{1}{3}\left[2Y_2^{(2)} Y_2^{(4)} - Y_3^{(2)} Y_1^{(4)} - Y_1^{(2)} Y_3^{(4)}\right]. \tag{A.9}$$

B Numerical fit procedure

`differential_evolution`: 3 seeds (42, 43, 44); maxiter= 150; popsize= 15; tol= 10^{-8}; polish=False, then Nelder–Mead polish (maxiter= 10000).

Bounds: $\mathrm{Re}\,\tau \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau_u \in [0.87, 2.5]$, $\mathrm{Im}\,\tau_d \in [0.87, 1.60]$, $c_i \in [10^{-8}, 10^2]$.

Observables: SVD $M_u = U_u^\dagger \Sigma_u V_u$, $M_d = U_d^\dagger \Sigma_d V_d$; masses $m_i = \Sigma_i$; CKM $V = U_u^\dagger U_d$; five mass ratios, $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$, $|J|$.

Runtime: 90–120 s per full run (Python 3.10 + scipy).

C Reproduction of τ and χ^2

Table 13: Three independent DE runs.

Source	$\mathrm{Re}\,\tau_u$	$\mathrm{Im}\,\tau_u$	$\mathrm{Re}\,\tau_d$	$\mathrm{Im}\,\tau_d$
Run 1	−0.15779	2.38518	+0.36939	1.54105
Run 2	−0.15779	2.38518	+0.36939	1.54105
Run 3	−0.12893	2.38411	−0.37212	1.54102

$\mathrm{Im}\,\tau_u = 2.385 \pm 0.001$; $\mathrm{Im}\,\tau_d = 1.541 \pm 0.0001$.

Rounding sensitivity:

Table 14: Sensitivity of χ^2 to τ rounding.

Precision of τ	χ^2
exact (10 digits)	6.195×10^{-20}
4 significant digits	1.08×10^{-3}
3 significant digits	2.55×10^{-3}

D Proof of coefficient degeneracy

At fixed τ (2.3), multiply all c_i by k:

Table 15: Invariance of χ^2 under $c_i \to kc_i$.

k	χ^2
0.5	6.20×10^{-20}
1.0	6.20×10^{-20}
1.87	1.08×10^{-19}
2.0	6.20×10^{-20}
5.0	3.00×10^{-19}
10.0	3.00×10^{-19}

All $\chi^2 < 10^{-18}$: $\chi^2(kc) \approx \chi^2(c)$. **Uniform scale degeneracy proven.**

E Flatness of $\mathrm{Re}\,\tau$

Scan in $\mathrm{Re}\,\tau_u$ (at $\mathrm{Re}\,\tau_d = +0.369$, coefficients fixed): width of $\chi^2 < 10$: $[-0.40, +0.00]$ (0.40).

Scan in $\mathrm{Re}\,\tau_d$**:** width of $\chi^2 < 10$: $[+0.30, +0.50]$ (0.20).

At fixed coefficients $\mathrm{Re}\,\tau$ is soft-flat. At refit (Appendix C) $\mathrm{Re}\,\tau_d \in [-0.37, +0.37]$.

F LOO test: procedure and results

For each $j \in \{1, \ldots, 9\}$: remove the j-th observable, fit $8 \to 7$ coefficients, predict the j-th. $N_{\mathrm{dof}} = +1$.

Results U6×D6: mean $|V_{ub}|$ +16728%, std 11819%; $|J|$ +3598%, std 7487%.

Direct relation to degeneracy (Appendix D).

G Systematic search for α_G

Basis: 16 elements ($Y_i^{(2)}$, $Y_i^{(4)}$, $Y_i^{(6)}$, η, π, $\sqrt{\pi}$, π^2, $e^{-\pi^2}$), where $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ is the Dedekind eta function. Tested $N = 392$ combinations: $|a|/|b|$ (240), $|a| \cdot |b|$ (120), $\sqrt{|a|/\pi}$ (16), $|a|^2/\pi^{3/2}$ (16).

2 coincidences below 0.1%**:** $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (-0.0385%, $n = 5..30$); $|Y_2^{(6)}|/\pi = 0.212411$ ($+0.0965\%$, $n = 5..30$).

Poisson: $\langle n \rangle = 392 \cdot 0.002 = 0.78$; $n_{\mathrm{obs}} = 2$; $P(n \geq 2) = 0.185$ (not significant).

Conclusion: α_G remains a postulate.

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